The instability of negative-mass Schwarzschild is a point-mass effect:

Exact thresholds for finite stars and a radially stable negative-mass boson star

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Abstract

Negative-mass Schwarzschild spacetime is linearly unstable: Gleiser and Dotti exhibited, for every multipole $\ell \geq 2$, an even-parity mode growing as $e^{\gamma t}$ with $\gamma = (\ell-1)\ell(\ell+1)(\ell+2)/(12|M|)$, and gave numerical evidence that it is the only one. It has been suggested that the instability does not depend on the singularity and would survive in a regular negative-mass star. We prove, for every $\ell \geq 2$, that the Dotti–Gleiser operator has exactly one negative eigenvalue. The proof uses the zero-energy partner solution $\phi_0 = 1/\psi_0$ implicit in their intertwining construction, through a ground-state identity and Glazman's lemma. The same identity yields an exact count of unstable modes for the exterior of a star of areal radius R whose interior imposes a Robin condition $\hat{\Psi}' = \kappa\hat{\Psi}$: $\#\{\gamma > 0\} = [\, R < r_s \,] + [\, \kappa < \kappa_c(R) \,]$, with $r_s = 6|M|/[(\ell+2)(\ell-1)]$ and κ_c in closed form. Consequently the vacuum exterior of any star with $R \geq r_*(\ell)$ ($r_* = 2.279|M|$ for $\ell = 2$) and a non-energy-injecting surface condition ($\kappa \geq 0$) supports no unstable even-parity mode, and the ℓ^4 growth of the point mass is capped by any finite radius. We then study a concrete regular negative-mass object, a sextic boson star whose stress tensor enters Einstein's equations with a minus sign. We derive its radial perturbation equations, verify them symbolically, and find the configuration used in a recent full numerical-relativity study to be radially stable; the family loses stability exactly where the Noether charge is extremal. Because the first law becomes $\mathrm{d}M = -\omega\,\mathrm{d}N$, the turning-point criterion holds in N but reverses sign when written in M. Accurate spherically symmetric nonlinear evolutions confirm both the stationarity of the stable stars to $t = 1500$ and the predicted growth rates of unstable ones.

1 Introduction

Bondi's 1957 analysis [1] showed that general relativity contains no logical obstruction to negative mass, provided inertial, passive and active mass share a sign, and that a pair of equal and opposite masses self-accelerates (the "runaway"). Two questions have dominated the subsequent discussion: whether negative mass is dynamically stable, and whether the runaway is physical. The second was recently addressed by full $3+1$ evolutions of a canonical boson star paired with a "phantom" boson star of negative ADM mass [3]. Here we address the first.

For the vacuum point mass the answer is known. Gibbons, Hartnoll and Ishibashi [4] proved stability of the odd-parity sector and, for a range of boundary conditions at the naked singularity, of the even sector. Gleiser and Dotti [5] then found exact, smooth, exponentially growing even-parity solutions and showed that the stability argument of Ref. [4] relies on a perturbation that is not continuous at an interior radius where the Zerilli variable is singular. Dotti and Gleiser [6]

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removed that "kinematic" singularity with an intertwining operator, obtained a regular operator $\hat{H}$ with a unique self-adjoint extension, and showed that the unstable mode is excited by generic data supported away from the singularity. Cardoso and Cavaglià [7] identified the modes as algebraically special. Whether the unstable mode is the only one was left as numerical evidence [6].

What a regular negative-mass source does to this instability has not been computed. In a recent review Dotti suggests that replacing the singularity by a small negative-mass distribution would leave the potential largely unchanged, so that "an unstable mode could be found anyway", and that "the singularity plays no role in the instability" [8]. Regular negative-mass configurations exist—perfect-fluid bubbles in de Sitter [9], scalar-field stars [3]—but stability analyses have been radial and restricted to special matter models [10, 11].

This paper makes three contributions.

1. We prove that $\hat{H}$ has exactly one negative eigenvalue for every $\ell \geq 2$ (Theorem 1), turning the numerical observation of Ref. [6] into a theorem.

2. For the exterior of a star of radius R with a Robin interface condition, we give an exact count of unstable modes for every ℓ (Theorem 2). The instability is a property of the region $r \lesssim 10|M|/\ell^2$: a star larger than $r_*(\ell)$ hides it unless its surface condition injects energy, and any finite star caps the growth rate. This quantifies, and for dilute stars contradicts, the expectation of Ref. [8].

3. We give the first linear stability analysis of a negative-ADM-mass boson star [3]. It is radially stable on the branch used in numerical relativity, with stability lost exactly at the extremum of the Noether charge.

The non-radial stability of the boson star, which requires computing the effective interface condition that its interior supplies, is left for future work (Sec. 6).

Units $G = c = 1$. Sections 2–4 use $|M| = 1$; Sec. 5 uses the scalar mass $m = 1$. All computations are reproducible from the accompanying repository.

2 The intertwined Zerilli operator

Write the negative-mass Schwarzschild metric as $\mathrm{d}s^2 = -f\mathrm{d}t^2 + f^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2$ with $f = 1 - 2M/r = 1 + 2/r$ ($M = -1$), and the tortoise coordinate $x = r - 2\ln(1 + r/2)$, so $x \in (0, \infty)$ with $x \simeq r^2/4$ at the singularity. Even-parity perturbations with multipole $\ell \geq 2$ reduce to Zerilli's equation [12] $\partial_t^2 \Psi_Z + H_Z \Psi_Z = 0$, $H_Z = -\partial_x^2 + V_Z$. Let

$$c = (\ell + 2)(\ell - 1), \qquad k = \frac{(\ell - 1)\ell(\ell + 1)(\ell + 2)}{6} = \frac{c(c + 2)}{6}, \qquad r_s = \frac{6}{c}. \tag{1}$$

For $M < 0$ the potential V_Z has a double pole at r_s, which lies in the physical domain; it is "kinematic" because Ψ_Z itself has a simple pole there for smooth metric perturbations [5].

Dotti and Gleiser [6] use the static solution $\psi_0 = \chi(r)/(cr - 6)$ of $H_Z \psi_0 = 0$, where χ is the polynomial of degree $\ell + 2$ in their Eq. (55), and define $g = \psi_0'/\psi_0$ ($' = \mathrm{d}/\mathrm{d}x$),

$$A = \partial_x + g, \qquad H_Z = AA^*, \qquad \hat{H} = A^*A = -\partial_x^2 + \hat{V}, \qquad \hat{V} = g^2 - g'. \tag{2}$$

$\hat{V}$ is regular, $\hat{H}A^* = A^* H_Z$, and $\hat{\Psi} = A^* \Psi_Z$ is smooth for smooth metric perturbations. The zero-energy partner of this factorisation,

$$\phi_0 = \frac{1}{\psi_0} = \frac{cr - 6}{\chi(r)}, \qquad A\phi_0 = 0, \qquad \hat{H}\phi_0 = 0, \tag{3}$$

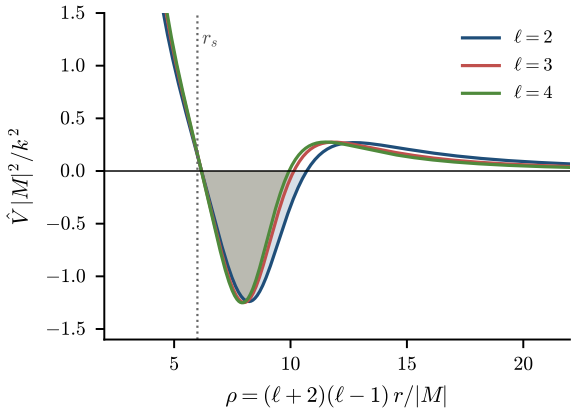


Figure 1: The regularised potential $\hat{V}$ for $\ell = 2, 3, 4$, in the scaled radius $\rho = cr/|M|$. Shaded: $\hat{V} < 0$, which occurs only on a shell just beyond r_s ($\rho = 6$). With radius scaled by $c = (\ell+2)(\ell-1)$ and the potential by the eigenvalue scale k^2, the curves nearly coincide.

appears in Ref. [6] as the state "missing in the spectrum" of the supersymmetric partner. It is the object on which everything below rests. Figure 1 shows $\hat{V}$: it is negative only on a thin shell just outside r_s, and when lengths are scaled by c and energies by k^2 the shell is nearly independent of ℓ.

Lemma 1 (properties of χ, ϕ_0, $\hat{V}$; all $\ell \geq 2$)**.**

1. $\chi = \sum_{n=1}^{\ell+2} c_n r^n$ *with* $c_1 = \frac{3}{2}$, $c_2 = 0$, $c_3 = -K/32$, $c_4 = K/96$ *($K = 6k$),* $c_n \geq 0$ *for* $n \geq 4$ *and* $c_{\ell+2} > 0$*;* ψ_0 *solves the static Zerilli equation identically in* ℓ*; and* $\chi > 0$ *on* $r > 0$*.*

2. *At* r_s *the* χ*-equation has Frobenius exponents* 0 *and* 3*, forcing* $\chi''(r_s) = 0$*; hence* $\hat{V}$ *is real-analytic on* $(0, \infty)$*.*

3. $r^4\hat{V} = 12 + O(r)$ *as* $r \to 0$ *(exponents* $x^{3/2}$*,* $x^{-1/2}$*), and* $\hat{V} \simeq (\ell+1)(\ell+2)/r^2$ *as* $r \to \infty$*.* $\hat{H}$ *is limit point at both ends, hence essentially self-adjoint on* $C_c^\infty(0, \infty)$*.*

4. ϕ_0 *is the non-admissible branch* $\propto x^{-1/2}$ *at* 0*, decays as* $r^{-(\ell+1)}$ *at infinity, and has exactly one zero, at* r_s*, which is simple.*

Sketch. (1) The closed form $c_n = (n-2)\big((n-4)\ell(\ell+1) + n - 1\big)(\ell - n + 3)_{2n-4}/\big(2^n(n-1)!^2\big)$ turns the static equation into a three-term recurrence, which becomes a polynomial identity in (ℓ, n) after dividing by a Pochhammer factor; it is verified symbolically. Positivity follows Ref. [6], Eqs. (57)–(59): dropping the non-negative terms $n \geq 6$ leaves a quartic whose unique positive critical value is $4(K-18)(Ks - 18(K-16))/[(\ell-2)(\ell+3)(\ell^2+\ell+4)]^3$, $s = \sqrt{6(K-18)}$, which is positive because $(Ks)^2 - (18(K-16))^2 = 6(K-24)^3 > 0$ for $\ell \geq 3$; $\ell = 2$ is checked directly. (2)–(4) are local expansions, verified symbolically in ℓ; limit point at 0 holds because $x^{-1/2} \notin L^2$, and the Weyl alternative gives essential self-adjointness [14, 15]. □

3 Exactly one unstable mode

The key tool is the ground-state identity: if $\hat{V} = \phi_0''/\phi_0$ and $u = \phi_0 h$, then

$$u'^2 + \hat{V}u^2 = \phi_0^2 h'^2 + \big(\phi_0\phi_0' h^2\big)'. \tag{4}$$

Theorem 1. *For every $\ell \geq 2$, $\hat{H}$ has exactly one negative eigenvalue. It is simple and equals $-(k/2)^2$, so the unique unstable even-parity mode of negative-mass Schwarzschild grows at the Gleiser–Dotti rate $\gamma = k/(2|M|)$. Zero is not an eigenvalue.*

Proof. At most one. By Glazman's lemma [16], the number of negative eigenvalues equals the largest dimension of a subspace of a form core on which $q[u] = \langle u, \hat{H}u\rangle$ is negative definite. Take the core $C_c^\infty(0,\infty)$ and suppose such a subspace F has $\dim F \geq 2$. Choose $u \in F \setminus \{0\}$ with $u(x_s) = 0$, $x_s = x(r_s)$. Since ϕ_0 is analytic with a simple zero at x_s, $h = u/\phi_0$ is smooth, and integrating (4) over the compact support of u gives $q[u] = \int \phi_0^2 h'^2 \,\mathrm{d}x \geq 0$, a contradiction.

At least one. The function $\hat{\Psi} = A^*\Psi_{\mathrm{DG}}$, with $\Psi_{\mathrm{DG}} = r(r+2)^k e^{-kr/2}/(cr-6)$, satisfies $\hat{H}\hat{\Psi} = -(k/2)^2\hat{\Psi}$ identically in ℓ, is $O(r^3)$ at 0, regular at r_s, and decays exponentially; so it is an eigenfunction. Independently of this explicit mode, $w = \phi_0 \mathbf{1}_{x>x_s}$ lies in the form domain with $q[w] = 0$ by (4); if $\hat{H} \geq 0$, w would be a minimiser and hence a weak, therefore C^1, solution of $\hat{H}w = 0$, contradicting the kink at x_s.

No zero mode. The zero-energy solution admissible at $x = 0$ is

$$v = \phi_0 F, \qquad F(r) = \text{f.p.}\int_0^r \psi_0^2 \,\frac{\mathrm{d}x}{\mathrm{d}r}\,\mathrm{d}r;$$

it grows as $r^{\ell+2}$ and is not in L^2. □

Remark 1. v has exactly one zero for every ℓ (F is positive on $(0, r_s)$ and increases from $-\infty$ to $+\infty$ on (r_s, ∞)), as renormalised oscillation theory [17] requires. The proof does not use oscillation theory at the continuum edge. Numerically, Chebyshev discretisations with $\hat{\Psi} = r^3 u$ reproduce $E_0 = -(k/2)^2$ to $\leq 3 \times 10^{-13}$ for $\ell \leq 8$ and find no second bound state in 140 resolution and domain variations.

4 Finite stars: an exact count

Model the exterior of a star of areal radius R by $\hat{H}_{R,\kappa}$ on $L^2(x_R, \infty)$ with the Robin condition $\hat{\Psi}'(x_R) = \kappa\hat{\Psi}(x_R)$; $\kappa = +\infty$ denotes Dirichlet. The interior matter determines κ, which in general depends on frequency and ℓ; here it is a free parameter. Define

$$\kappa_c(R) = \frac{\mathrm{d}\ln\phi_0}{\mathrm{d}x}\Big|_R = \Big(1 + \frac{2}{R}\Big)\frac{\mathcal{N}(R)}{(cR-6)\chi(R)}, \qquad \mathcal{N} = c\chi - (cr-6)\chi'. \tag{5}$$

Theorem 2. *For every $\ell \geq 2$ and $R \neq r_s$, the number of negative eigenvalues of $\hat{H}_{R,\kappa}$ is*

$$[\,R < r_s\,] + [\,\kappa < \kappa_c(R)\,], \tag{6}$$

with Dirichlet counted as $\kappa = +\infty$. At $\kappa = \kappa_c$, ϕ_0 is a zero-energy eigenfunction.

Proof. For u in the core, (4) gives $q[u] = \int \phi_0^2 h'^2 \,\mathrm{d}x + (\kappa - \kappa_c)u(x_R)^2$ provided $u(x_s) = 0$ when $R < r_s$. Imposing $u(x_s) = 0$ (if $R < r_s$) and $u(x_R) = 0$ (if $\kappa < \kappa_c$) costs at most $[\,R < r_s\,] + [\,\kappa < \kappa_c\,]$ linear conditions and leaves $q \geq 0$; Glazman's lemma gives the upper bound. For the lower bound use $w_1 = \phi_0 \mathbf{1}_{[x_R, x_s]}$ and $w_2 = \phi_0 \mathbf{1}_{[x_s,\infty)}$ (or ϕ_0 itself if $R > r_s$), for which $q[aw_1 + bw_2] = a^2\phi_0(x_R)^2(\kappa - \kappa_c)$ and $q[w_2] = 0$. If the negative spectral subspace were smaller, some combination orthogonal to it would have $q = 0$, hence be a weak zero-energy solution satisfying the boundary condition; the kink of w_2, or $\kappa \neq \kappa_c$, excludes this. □

ℓ	r_s	r_*	$r_{\rm neg}$	$\gamma_{\rm DG} = k/2$	$\lvert M\rvert/r_*$
2	1.5000	2.2788	2.6741	2	0.439
3	0.6000	0.8789	1.0143	10	1.14
4	0.3333	0.4814	0.5519	30	2.08
5	0.2143	0.3073	0.3512	70	3.25

Table 1: Thresholds in units of $|M|$. r_s: Dirichlet threshold; r_*: Neumann and $\kappa \geq 0$ threshold; $r_{\rm neg} = \sup\{r : \hat{V} < 0\}$. Since r_* decreases with ℓ (checked for $\ell \leq 80$), a star with $|M|/R < 0.439$ and $\kappa \geq 0$ has no unstable even-parity mode for any ℓ. At large ℓ, $r_* \approx 8.486/c$ and $r_{\rm neg} \approx 9.636/c$ (derived from the $\ell \to \infty$ limit of $c\,\chi(\rho/c)$, a modified-Bessel closed form; convergence of the roots is not proven).

Corollary 1 (thresholds, all ℓ)**.** *$\mathcal{N} > 0$ on $[0, r_s]$ and has exactly one root $r_* > r_s$. Hence $\kappa_c < 0$ on $(0, r_s)$, $\kappa_c \to \mp\infty$ at $r_s^{\mp}$, $\kappa_c > 0$ on (r_s, r_*) and $\kappa_c < 0$ on (r_*, ∞). In particular: Dirichlet is unstable iff $R < r_s$; Neumann is unstable iff $R < r_*$; every $\kappa \geq 0$ is stable iff $R \geq r_*$.*

The first claim follows from a telescoping lower bound on $[0, r_s]$; uniqueness of r_* from Descartes' rule applied to $\mathcal{N}(r_s+t)$, whose coefficients have one sign change because $\chi''(r_s) = 0$ and $\chi^{(m)}(r_s) > 0$ for $m \geq 4$. Table 1 lists the thresholds and Fig. 2 the critical surface condition. Asymptotically,

$$\kappa_c = -\frac{\ell+1}{R} - \frac{\ell^4 + 3\ell^3 + 4\ell^2 + 4}{\ell(\ell-1)(\ell+2)}\frac{1}{R^2} + O(R^{-3}) \qquad (R \to \infty), \tag{7}$$

$$\kappa_c\, x_R = -\frac{1}{2} - \frac{(\ell^2 + \ell - 1)R}{12} + O(R^2) \qquad (R \to 0). \tag{8}$$

For large R an instability therefore requires a surface condition softer than the static vacuum tail $\phi_0 \propto r^{-(\ell+1)}$ by an $O(1)$ factor on the scale $1/R$. As $R \to 0$ every condition with $\kappa x_R > -\frac{1}{2}$ (including Dirichlet and Neumann) reproduces the Dotti–Gleiser eigenvalue, with error scaling roughly as R^3–R^4, so the self-adjoint extension of Ref. [6] is the limit of a regular cutoff; compare the scalar-wave analysis of Ref. [13].

Growth-rate cap. For the point mass the growth rate $k/2 \sim \ell^4/12$ is unbounded in ℓ. For a star: with $\kappa \geq 0$ only ℓ with $R < r_*(\ell)$, roughly $\ell \lesssim 3/\sqrt{R}$, are unstable; with a fixed $\kappa < 0$, every ℓ with $r_{\rm neg}(\ell) < R$ has $\hat{V} \geq 0$ outside the star and growth $\leq |\kappa|$. Either way $\sup_\ell \gamma < \infty$ (Fig. 2, right; a 1000-case numerical scan agrees with Theorem 2). The unbounded growth belongs to the singularity.

5 A regular negative-mass star

Theorem 2 reduces the even-parity stability of any star with $R > r_*$ to the sign of the effective κ its interior supplies, together with the stability of the matter itself. We now examine the matter for a concrete model.

Model. Following Ref. [3], take a complex scalar with $\mathcal{L} = -\frac{1}{2}|\nabla\Phi|^2 - V(|\Phi|^2)$, $V(s) = \frac{1}{2}s - \frac{1}{4}\lambda s^2 + \frac{1}{6}\mu s^3$, $\lambda = 10240$, $\lambda^2/\mu = 4.8$, coupled through

$$G_{ab} = 8\pi\varepsilon\, T_{ab}[\Phi], \qquad \Box\Phi = 2V'(|\Phi|^2)\Phi, \tag{9}$$

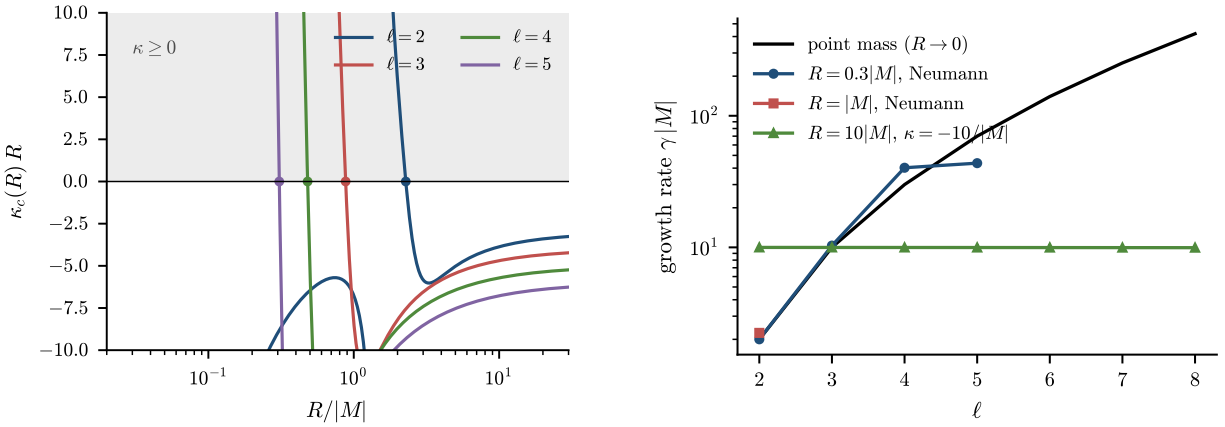


Figure 2: Left: critical surface condition $\kappa_c(R)R$ for $\ell = 2$–5. A mode is unstable when $\kappa < \kappa_c$, plus one unconditional mode when $R < r_s$ (pole). Dots mark r_*. Above r_* the whole $\kappa \geq 0$ region (shaded) is stable. Right: growth rate versus ℓ. The point mass grows as $\ell^4/12$; any finite R with an ℓ-independent condition leaves finitely many unstable ℓ ($\kappa \geq 0$) or caps the rate at $|\kappa|$ ($\kappa < 0$, since $\hat{V} \geq 0$ outside $r_{\rm neg}$).

with $\varepsilon = +1$ (canonical) or $\varepsilon = -1$. For $\varepsilon = -1$ the field equation is unchanged but the star gravitates with negative ADM mass. In the language of Bondi, the scalar has a single mass parameter that is inertial, passive and active at once, and is negative for $\varepsilon = -1$; gravity alone cannot distinguish this from Bondi's "all three negative" [1, 2]. The action is that of a ghost, so the model is a classical probe; quantum mechanically it is unstable to vacuum decay [18].

Static stars. Stationary solutions $\Phi = \phi(r)e^{-i\omega t}$ in Schwarzschild-like coordinates are solved as two-point boundary-value problems with a decaying Yukawa tail. The configuration of Ref. [3] at $\omega = 0.7603$ has $M = -0.014580$, rms radius 4.40 and $|M|/R \approx 3 \times 10^{-3}$, far below the threshold $|M|/R < 0.439$ of Table 1: the shell where negative-mass Schwarzschild would be unstable lies deep inside the matter. (Our masses are ADM masses; those quoted in Ref. [3] are the mass enclosed where ϕ falls to 5% of its central value and are 1.5–2.3% smaller; the field profiles agree to 10^{-5}.)

First law. With $N = \int 4\pi r^2 \omega \phi^2 (a/\alpha)\,\mathrm{d}r$ the Noether charge, the family satisfies

$$\mathrm{d}M = \varepsilon\,\omega\,\mathrm{d}N, \tag{10}$$

verified by finite differences to $< 10^{-4}$. Hence $M(\omega)$ and $N(\omega)$ are extremal at the same ω for either sign, but $\mathrm{d}M/\mathrm{d}\omega$ and $\mathrm{d}N/\mathrm{d}\omega$ have opposite signs when $\varepsilon = -1$.

Radial perturbations. Perturbing the metric functions and both components of the field with $e^{i\sigma t}$, we eliminate the metric through the tr and rr constraints and obtain a closed fourth-order system for the real perturbation, a gauge-invariant phase and the perturbed enclosed charge, containing σ only as σ^2 and ε symbolically. Substituting back, all six linearised field equations are satisfied identically. The structure is that of Refs. [19, 20]. Eigenvalues are found by compound-matrix shooting, which keeps the two-dimensional space of decaying solutions well conditioned, and are cross-checked by an independent finite-difference pencil that also returns complex eigenvalues.

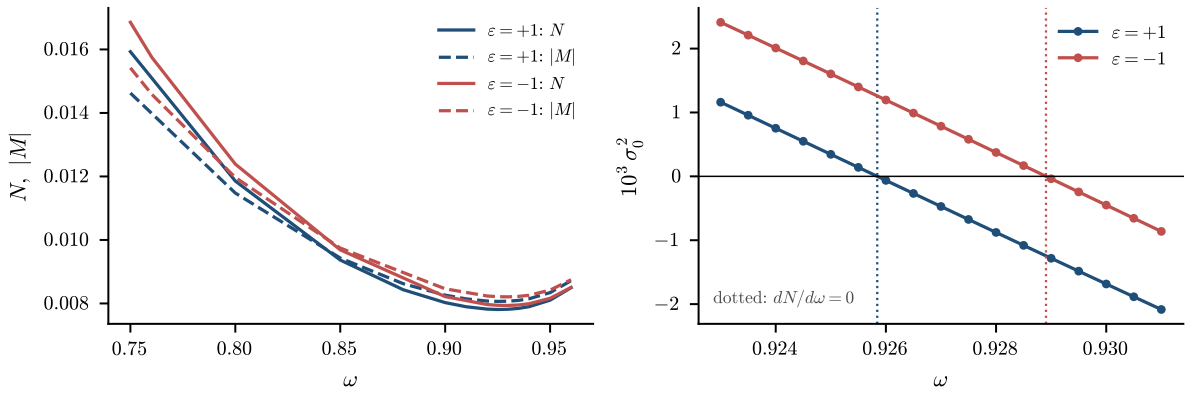


Figure 3: Radial stability of signed-source boson stars. Left: Noether charge N and $|M|$ along the canonical ($\varepsilon = +1$) and phantom ($\varepsilon = -1$) families. Right: fundamental σ_0^2 near the stability change; dotted lines mark $\mathrm{d}N/\mathrm{d}\omega = 0$, which coincides with $\sigma_0^2 = 0$ to 2×10^{-8} ($\varepsilon = +1$) and 8×10^{-8} ($\varepsilon = -1$) in ω.

Results. For $\varepsilon = +1$ the textbook criterion is reproduced at both turning points of the family: the fundamental mode crosses zero at $\omega = 0.92584444$ against $\mathrm{d}M/\mathrm{d}\omega = \mathrm{d}N/\mathrm{d}\omega = 0$ at 0.92584446, and again at the strong-field maximum of N ($\omega \approx 0.200$, $N \approx 42.73$) within the scan resolution. For $\varepsilon = -1$ (Fig. 3):

- The star of Ref. [3] is radially stable: no real growth rate up to $|\sigma| = 2$ and no complex eigenvalue with $|\sigma^2| < 0.3$, at two resolutions of the finite-difference check.
- The family is stable for $\omega < 0.92891$ and unstable above. The crossing $\sigma_0^2 = 0$ at $\omega = 0.92890846$ coincides with $\mathrm{d}N/\mathrm{d}\omega = 0$ at 0.92890838.
- The criterion is therefore invariant when stated in N (stable side $\mathrm{d}N/\mathrm{d}\omega < 0$) but reverses when stated in M: on the stable side $\mathrm{d}M/\mathrm{d}\omega = -\omega\,\mathrm{d}N/\mathrm{d}\omega > 0$.
- The phantom family has a fold in ω ($\omega_{\min} \approx 0.4223$) at which neither M nor N is extremal; the fundamental mode stays positive through it. The strong-field branch, followed to $|2m/r| \approx 0.4$, is stable with no turning point.

Near $\omega \approx 0.93$ the instability is essentially the flat-space Vakhitov–Kolokolov instability of Q-balls [21, 22], with gravity a small correction; the strong-field branch is the more sensitive test of how ε enters. The first law and the N-form of the criterion follow from the standard turning-point arguments [23–25] with ε kept; the new content is the explicit linear analysis for $\varepsilon = -1$. The result is consistent with the theorem that barotropic negative-mass stars are radially unstable [11], which does not apply to scalar-field matter.

Nonlinear check. We evolve single stars in spherical symmetry in polar-areal slicing, re-solving the Hamiltonian constraint for the Misner–Sharp mass [26] and the slicing condition at every substep, with ε appearing only there [27, 28]. The scheme is fourth order in space and time and converges at order ≈ 4; the Misner–Sharp mass outside the star equals the ADM mass of the static solution to 5×10^{-8}. Unstable stars at $\omega = 0.94$, perturbed by $\pm\delta$, grow at $\sigma = 0.07376$ ($\varepsilon = +1$) and 0.06679 ($\varepsilon = -1$) against 0.07370 and 0.06668 from the linear analysis (Fig. 4, right). The

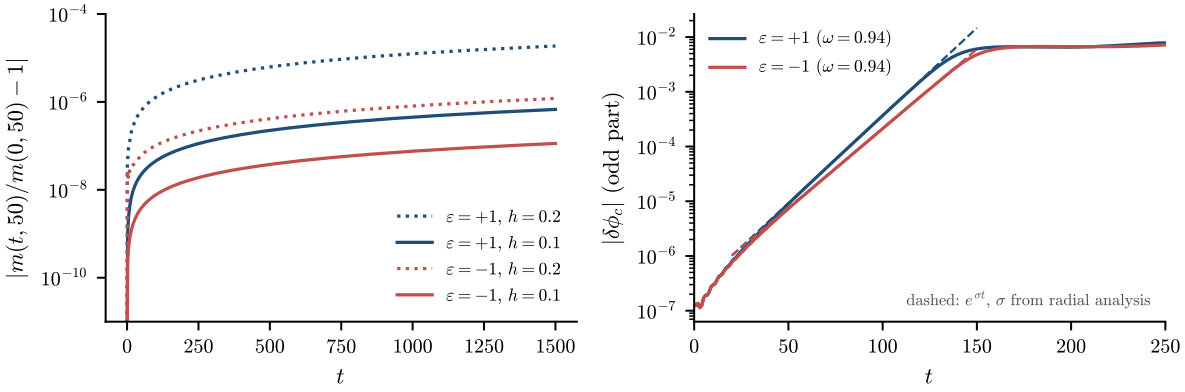


Figure 4: Spherically symmetric nonlinear evolutions. Left: relative drift of the Misner–Sharp mass at $r = 50$ for the stable campaign stars ($\omega = 0.750$ canonical, $\omega = 0.7603$ phantom) at two resolutions; the drift is a dissipation-limited truncation error that decreases rapidly with h. Right: odd part of the central-amplitude response of unstable stars ($\omega = 0.94$) to $\pm 10^{-5}$ perturbations; dashed lines have the growth rates of the linear analysis.

stable campaign stars of both signs remain stationary to $t = 1500$, with fractional drifts of ω, the central amplitude, the Misner–Sharp mass and N all $\leq 10^{-6}$ at $h = 0.1$ and decreasing with resolution (Fig. 4, left). A 5% perturbation relaxes within ~ 150 time units onto the neighbouring static star and stays there. The slow drifts reported for a lone phantom star in 3+1 [3] are therefore not a property of the spherical dynamics.

6 Discussion

For the point mass, Theorem 1 completes the linear stability analysis of negative-mass Schwarzschild: one unstable even-parity mode per ℓ, at the Gleiser–Dotti rate, and no others. Theorem 2 shows where that mode lives. The negative part of the effective potential sits on a shell of radius $\approx 10|M|/\ell^2$ (Fig. 1); a star whose surface lies outside $r_*(\ell)$ hides it unless its interior supplies an energy-injecting surface condition, and for $\ell = 2$ the condition is $|M|/R < 0.439$. In this sense the instability is a point-mass effect, contrary to the suggestion in Ref. [8]: for a dilute regular source the vacuum exterior cannot drive it, and the ℓ^4 growth disappears.

What remains is a property of the matter. For the boson star of Ref. [3], which is dilute ($|M|/R \sim 10^{-3}$), we have shown radial stability. Non-radial stability requires the effective interface condition $\kappa_{\rm eff}(\ell, \omega)$ produced by the interior, together with any instability intrinsic to the matter; the polar perturbation equations of the signed-source boson star are the natural route, and we leave them for future work. Theorem 2 reduces the gravitational part of that question to a comparison of $\kappa_{\rm eff}$ with the closed-form $\kappa_c(R)$.

Two limitations should be stated. Our results are classical: the phantom sector is a ghost, and a quantum theory of it has an unstable vacuum [18]. And the Robin model of the interior is exact only for an interface condition independent of frequency; a real star supplies a frequency-dependent one, for which Theorem 2 gives the count at each fixed frequency.

Reproducibility. The symbolic verifications (Lemma 1, Theorems 1–2, exact for $\ell = 2$–10 and symbolically in ℓ where stated; root isolation of $\mathcal{N}$ for $\ell \leq 50$), the spectral computations, the

boson-star background and perturbation codes, the nonlinear evolution code, and the script that generates every figure are in the accompanying repository.

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